

1) Circular measure

a) Radians

Example 1

Convert 165° to radians.

$$165 \times \frac{\pi}{180} = \frac{11\pi}{12} = 2.88 \text{ radians (3 s.f.)}$$

Multiply by $\frac{\pi}{180}$.

Example 2

Convert $\frac{2\pi}{3}$ radians to degrees.

$$\frac{2\pi}{3} \times \frac{180}{\pi} = 120^\circ$$

Multiply by $\frac{180}{\pi}$.

Or

$$\frac{2\pi}{3} = 2 \times \frac{\pi}{3} = 2 \times 60^\circ$$

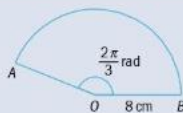
Note: It is a useful skill to be able to find multiples of known equivalents quickly in your head.

b) Arc length and sector

Example 3

OAB is a sector of a circle with centre O and radius 8 cm.

- Find the area of the sector in terms of π .
- Find the perimeter of the sector in terms of π .



$$\text{a) } A = \frac{1}{2}r^2\theta = \frac{1}{2} \times 8^2 \times \frac{2\pi}{3} = \frac{64\pi}{3} \text{ cm}^2$$

$$\text{b) } s = r\theta = 8 \times \frac{2\pi}{3} = \frac{16\pi}{3}$$

$$\text{Perimeter} = \text{arc length} + 8 + 8 = \frac{16\pi}{3} + 16 \text{ cm}$$

Substitute $r = 8$ and $\theta = \frac{2\pi}{3}$ into the formulae.

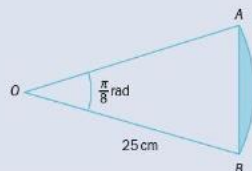
Example 4

OAB is a sector of a circle with centre O and radius 25 cm.

The line AB is a **chord** of the circle.

Find the area of the shaded region.

(You are given that the area of triangle OAB may be found by using the formula $\frac{1}{2} \times OA \times OB \times \sin AOB$.)



$$\text{Area of shaded region} = \text{Area of sector } OAB - \text{Area of triangle } OAB$$

$$\text{Area of sector } OAB = \frac{1}{2}r^2\theta = \frac{1}{2} \times 25^2 \times \frac{\pi}{8}$$

Find the area of the sector.

$$\text{Area of triangle } OAB = \frac{1}{2} \times 25 \times 25 \times \sin \frac{\pi}{8}$$

Find the area of the triangle.

$$\text{Area of the shaded region} = \frac{1}{2} \times 25^2 \times \frac{\pi}{8} - \frac{1}{2} \times 25^2 \times \sin \frac{\pi}{8}$$

$$= 3.13 \text{ cm}^2 \text{ (3 s.f.)}$$



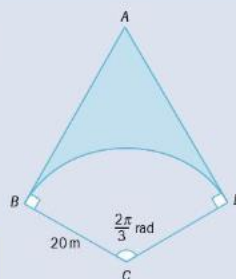
You should use the radian mode in your scientific calculator.

Example 5

In the diagram, BD is the arc of a circle with centre C and radius 20 m.

Angle $BCD = \frac{2\pi}{3}$ radians.

AB and AD are tangents to the circle at B and D , respectively.



a) Show that $AB = 20\sqrt{3}$ cm.

b) Show that the area of the shaded region is $400\left(\sqrt{3} - \frac{\pi}{3}\right)$ cm².

c) Find the perimeter of the shaded region in terms of π and $\sqrt{3}$.

Note: You may assume that $\tan \frac{\pi}{3} = \sqrt{3}$ in this question. Exact trigonometric ratios will be studied in Chapter 5.

a) Angle $BCA = \frac{\pi}{3}$

$$\tan \frac{\pi}{3} = \frac{AB}{20}$$

$$AB = 20 \tan \frac{\pi}{3}$$

$$AB = 20\sqrt{3} \text{ cm}$$

Use the right-angled triangle ABC .

Find AB .

b) Shaded area = $2 \times$ area of triangle ABC – area of sector DCB

$$= 2 \times \frac{1}{2} \times 20\sqrt{3} \times 20 - \frac{1}{2} \times 20^2 \times \frac{2\pi}{3}$$

$$= 400\sqrt{3} - \frac{400\pi}{3}$$

$$= 400\left(\sqrt{3} - \frac{\pi}{3}\right) \text{ cm}^2$$

Use the formulae: area of a triangle = $\frac{1}{2}bh$
and area of a sector = $\frac{1}{2}r^2\theta$.

c) Perimeter = $AB +$ arc length $BD + AD$

$$= 20\sqrt{3} + 20 \times \frac{2\pi}{3} + 20\sqrt{3}$$

$$= 40\sqrt{3} + \frac{40\pi}{3} \text{ cm}$$

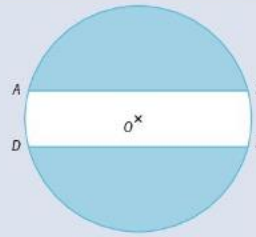
Use arc length = $r\theta$ and $AD = AB$.

Leave your answer in exact form.

Example 6

The diagram shows a shape $ABCD$ made by removing the two shaded regions from the circle of radius 8 cm and with centre O .

AB is parallel to DC . $AB = DC = 14$ cm.



- Find angle AOB . Give your answer in radians correct to 4 significant figures.
- Find the perimeter of $ABCD$.
- Find the area of $ABCD$.

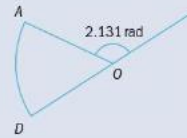
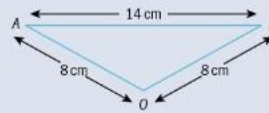
- a) Using the cosine rule:

$$\begin{aligned}\cos AOB &= \frac{8^2 + 8^2 - 14^2}{2 \times 8 \times 8} \\ &= \frac{-17}{32}\end{aligned}$$

Recall the cosine rule:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Angle } AOB = 2.131 \text{ rad (4 s.f.)}$$



- b) Perimeter of $ABCD = 2 \times \text{arc length } AD + 2 \times 14$

$$\text{Angle } AOD = \pi - 2.131$$

$$= 1.011 \text{ rad}$$

$$\text{Perimeter} = (2 \times 8 \times 1.011) + (2 \times 14)$$

$$= 44.2 \text{ cm (3 s.f.)}$$

Give your answer to 3 significant figures.

- c) Area of triangle $AOB = \frac{1}{2} \times 8 \times 8 \times \sin 2.131 = 27.11$

$$\text{Area of sector } OAD = \frac{1}{2} \times 8^2 \times 1.011 = 32.35$$

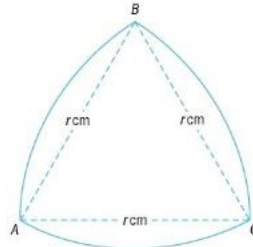
$$\text{Area of } ABCD = 2 \times 27.11 + 2 \times 32.35$$

$$= 119 \text{ cm}^2 \text{ (3 s.f.)}$$

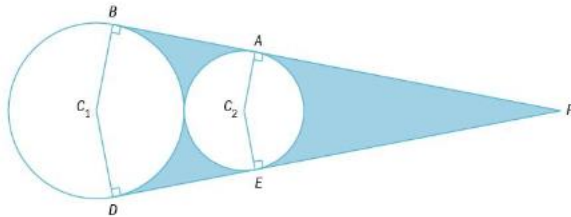
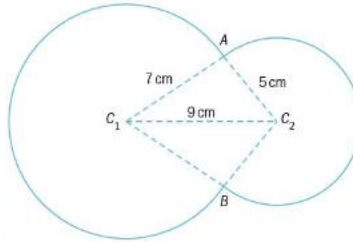
2 × area of triangle + 2 × area of sector

Use $\frac{1}{2}ab \sin C$ and $\frac{1}{2}r^2\theta$.

4. The diagram shows a shape ABC .
The boundary of the shape is made from three arcs.
The arcs AB , BC and AC have centres C , A , B , respectively.
- Find the perimeter of the shape in terms of r and π .
 - Find the area of the shape when $r = 15$ cm.



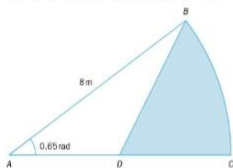
5. A metal plate is made from two circles.
One circle is of radius 7 cm.
The other circle is of radius 5 cm.
Their centres C_1 and C_2 are 9 cm apart.
- Find, in radians, angles AC_1B and AC_2B .
 - Find the area of the metal plate.
 - Find the perimeter of the metal plate.
6. In the diagram below, PAB and PED are tangents to two circles which touch and have centres C_1 and C_2 , respectively and radii 10 cm and 6 cm, respectively.
- Calculate the angle APE in radians.
 - Calculate the total area of the shaded regions.
 - Calculate the total of the perimeters of the shaded regions.



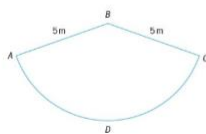
Summary exercise 4

EXAM-STYLE QUESTIONS

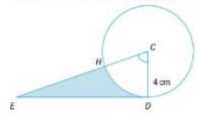
1. In the diagram, ABC is a sector of a circle with centre A and radius 8 m. The size of angle BAC is 0.65 rad and D is the mid-point of AC . Find
- the length of the arc BC
 - the area of the shaded region BCD .



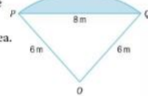
2. In the diagram below, $ABCD$ is a sector of a circle with centre B and radius 5 m. Given that the length of the arc ADC is $\frac{15\pi}{4}$ m, find
- the exact size of angle ABC in radians
 - the exact area of the sector $ABCD$.



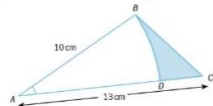
3. A circle has centre C and radius 4 cm. The diagram shows the circle with a point D which lies on the circle. The tangent at D passes through the point E . $EC = \sqrt{65}$ cm. Find
- the size in radians of angle DCE
 - the area of the shaded region DEH .



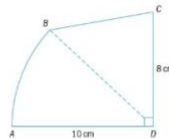
4. In the diagram below, OPQ is a sector of a circle, radius 6 m. The straight line PQ is 8 m long. Calculate
- the area of the sector OPQ
 - the shaded area.



5. The diagram below shows a triangle ABC with $AB = 10$ cm, $AC = 13$ cm and angle $BAC = 0.5$ rad. BD is an arc of a circle with centre A and radius 10 cm. Find
- the length of arc BD
 - the perimeter of the shaded region
 - the area of the shaded region.

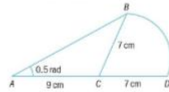


6. The diagram shows a shape $ABCD$. The curve AB is an arc of a circle with centre D . Angle ADB is 0.75 rad. Find
- the length of the arc AB
 - the area of the sector ADB
 - the area of $ABCD$.

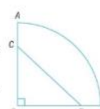


EXAM-STYLE QUESTION

7. A shape is made from joining a triangle ABC to a sector CBD of a circle with radius 7 cm and centre C . The points A , C and D lie on a straight line with $AC = 9$ cm. Angle $BAC = 0.5$ radians. Find
- the angle ACB giving your answer in radians
 - the area of $ABCD$.

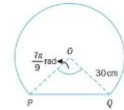


8. The diagram shows the sector OAB of a circle with centre O and radius 20 m. Angle $AOB = \frac{\pi}{2}$ radians. The straight line CD divides the sector into two regions which have equal areas and the lengths of OC and OD are equal. Calculate the exact length of OC .



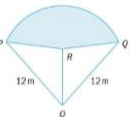
EXAM-STYLE QUESTION

9. The diagram shows a shape made from the arc of a circle with centre O and radius 30 cm and the straight line PQ . The size of angle POQ is $\frac{7\pi}{9}$ radians.

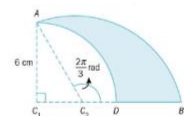


Find the total perimeter of the shape.

10. The diagram shows a sector OPQ of a circle with centre O and radius 12 m. $PR = QR = OR = 8$ m. Work out the area of the shaded region PQR .



11. AD is an arc of a circle with centre C_1 and radius 6 cm. AB is an arc of a circle with centre C_2 . The size of angle AC_2D is $\frac{2\pi}{3}$ radians.



Calculate

- the radius of the circle with centre C_2
- the area of the shaded region ABD .

Summary:

Radians and degrees

- $360^\circ = 2\pi$ radians.
- To convert from degrees to radians, multiply by $\frac{\pi}{180}$.
- To convert from radians to degrees, multiply by $\frac{180}{\pi}$.

Degrees	0°	30°	45°	60°	90°	180°	270°	360°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π

Arc length and area of a sector

- The angle for a whole circle = 2π radians.
- Arc length = $r\theta$ (θ measured in radians).
- Area of sector = $\frac{1}{2}r^2\theta$ (where θ is measured in radians and r is the radius of the circle).